

Predicting a Successful Climb of Mount Everest

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Abstract

Abstract to be written later

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Introduction

Mount Everest is the highest land surface point on the planet, when this height is defined as height above mean sea level. At 8,848 m (29,031.7 feet), it towers above its closest rival by more than 200 m (See list of highest mountains on Earth:

https://en.wikipedia.org/wiki/List_of_highest_mountains_on_Earth). While there are more than 100 peaks on the planet with peaks above 7 km, none captures the human imagination as Everest. Humans cannot live at such high elevations far above the tree line, where the air is rarefied and every function of the human body is challenged Goos, 2018.

Despite the extreme environment of peaks about 7,000 m, the human desire to overcome a challenge even at extremely personal risk remains alive and well. Mountains are not the only extreme environment on the planet that humans have tried to reach. The deepest parts of the oceans are also areas we have successfully traversed, however, the costs of specialized equipment and availability of aid in achieving the goal of reaching these extreme environment has made Everest an attractive place because it became attainable and relatively inexpensive.

For centuries now, climbing the world's highest mountains has been enabled and facilitated by modern technology and, in the case of Everest, Nepal tourist industry built around climbing the mountain. These two factors have made one of the world's deadliest mountains accessible to even more adventurers, and many would say, amateurs. In the early days of climbing teams climb to be the first to reach the highest peak, now anyone with enough resources can travel to, especially Everest and attempt to summit. But far from all who do, make it, or even make it back in one piece - if they make it back at all.

Everest and K2 (second peak in Karakoram) have over the years competed for the title: deadliest mountain. Many lives have been lost due a range of factors ranging from physiology, mindset, preparedness, ego, stupidity and natural disasters such as avalanches and inclement weather.

Many studies and documentaries have been done on mountain climbers and their plight to summit and return - alive. But sadly, the history of climbing is replete with tragedies. In the light of this

some articles have tried to analyze and suggest ways to improve your chances of a successful summit. In this paper we define that success as one where the climber summits and returns under their own strength, and of course, alive! For example, Huey et al. (2001) analyzed a research program that attempted to predict (similar to this paper) what factors influence the success ascent or death rates. In this project we do not try to predict death rates as not all failures to summit is due to the mountaineer dying. However, the paper drew no conclusions on the matter. Szymczak et al. (2021) examined the weather climber experienced on Everest and K2 in both the climbing season (July) and mid winter season. Their study found that more extreme weather in both seasons than on K2, which is 8° latitude farther north.

EDA

Null Hypothesis

Participant information goes here.

Materials

Model selection

Some words about the measures used.

Features and Ranking

Some more words

Results

Description of the procedure.

Conclusion

Placeholder

Table 1*Results of Exploratory Data Analysis*

Variable Name	Total Unique Values	Mean (μ)	STD (σ)	Continuous/Discrete
peak_name	385	-	-	D
nationality	97	-	-	D
year	93	2001.53	14.76	D
season	5	-	-	D
total_days	93	20.28	17.66	C
exp_result	15	-	-	D
max_elev_reached	696	7,106.91	1,763.61	C
mbrs_deaths	8	0.08	0.37	C
is_o2_not_used	2	0.71	0.46	D
is_o2_climbing	2	0.25	0.43	D
is_o2_descent	2	0.01	0.12	D
is_o2_medical	2	0.03	0.17	D
is_o2_unkwn	2	0.00	0.05	D
had_o2	0.09	2	0.28	D
accidents	2691	-	-	D

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